

ASSIGNMENT 4

Hands-on Subnetting Practical:

Task (1):

Dividing 192.168.10.0/24 into 4 equal subnets

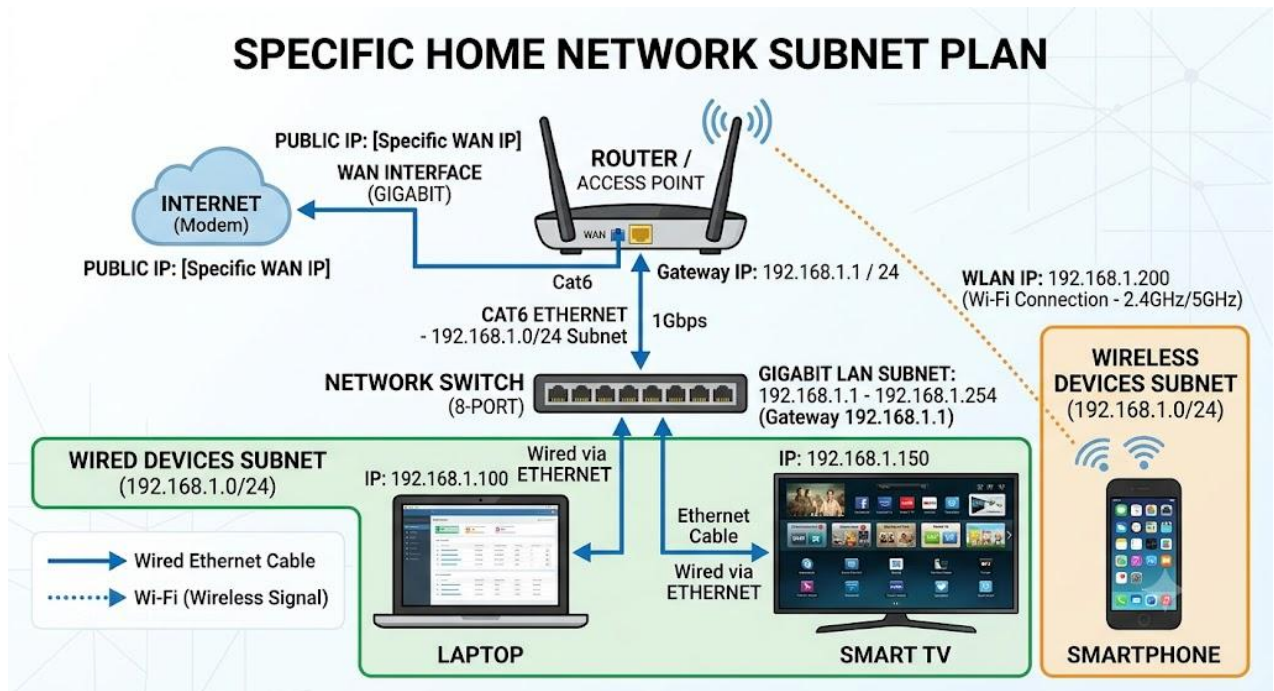
/24 -> /26

Subnet	Network Address	Subnet Mask	Broadcast Address
1	192.168.10.0/26	255.255.255.192	192.168.10.63
2	192.168.10.64/26	255.255.255.192	192.168.10.127
3	192.168.10.128/26	255.255.255.192	192.168.10.191
4	192.168.10.192/26	255.255.255.192	192.168.10.255

Task (2):

Device	Assigned IP Address	Subnet Mask
Router	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile Phone	192.168.10.5	255.255.255.192

Task (3):



Task (4):

- **Step 1: Choose the subnet**
 - We used **192.168.10.0/26**.
- **Step 2: Find the subnet mask**
 - **/26** means the subnet mask is:
 - **255.255.255.192**
- **Step 3: Find the number of hosts**
 - There are 6 host bits left:
 - $2^6 = 64$

➤ $64 - 2 = 62$ usable hosts

● **Step 4: Find the valid IP range**

➤ Network address: 192.168.10.0

➤ Broadcast address:

192.168.10.63

➤ Valid hosts: 192.168.10.1 -

192.168.10.62

● **Step 5: Assign IP addresses**

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5